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Evaluating the Added Value of Visible Structural-Defect Information for Pre-Storm Urban Tree Prioritization

Research Proposal

International Academic Research Competition (IARCO) 2026

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Introduction

Urban trees occupy a substantial part of the environment people live in and provide benefits such as shade, cooling, and stormwater interception. In New York City (NYC) alone there are over 7 million of them. Although trees benefit our lives, they might be the cause of secondary hazards after natural disasters as they are deliberately located near streets, homes, pedestrians, vehicles, and infrastructure, and their weight can be detrimental to facilities, utilities, and people once their structure fails during extreme events. During Tropical Storm Isaias in 2020, more than 21,000 tree-damage reports were generated in NYC. This raises an important question: how can cities use limited pre-storm inspection resources to identify trees that warrant the greatest attention before severe weather? [8], [9]

Cities manage a huge number of trees; thus, they have limited time and resources for intensive pre-storm inspections. They often possess basic inventory information such as species, DBH and condition, while more detailed assessments can record structural defects such as decay, deadwood, cracks and weak unions. However, the additional operational value of these defect observations for pre-storm prioritization remains insufficiently quantified.

Research question and Hypotheses

The study addresses a pressing practical challenge for the city. The research question: *How much does visible structural-defect information improve pre-storm prioritization of urban trees compared with routine inventory data alone under a limited inspection capacity?* This question becomes relevant since cities cannot inspect every tree in detail before a storm; thus, it is important to determine if investing extra time and budget to collect detailed deficiency data genuinely yields better prioritization choices.

To test this, two distinct independent variables are used: routine inventory data alone and combined routine inventory data with visible structural-defect information. Then, to measure their real-world performance there are three dependent variables: the prioritization performance measured by the proportion of storm-damaged trees identified within fixed inspection capacities 5%, 10%, and 20%.

Hypotheses:
Primary hypothesis: Adding visible structural-defect information to routine inventory data will increase the proportion of storm-damaged trees identified within a fixed pre-storm inspection budget.

Second hypothesis: The improvement from added visible structural-defect information will remain across different inspection budgets, such as 5%, 10%, and 20%.

Third hypothesis: The prioritization improvement will remain when the analysis excludes trees with earlier service request-triggered detailed inspections.

Literature review

Theme 1

Research on urban tree failure has mainly focused on identifying factors associated with failure, evaluating visual risk assessments, and developing models to predict which trees are more prone to fail. Kabir et al. (2018), Kane (2008), and van Haaften et al. (2021) examined tree characteristics associated with failure across different storm contexts. Their work points to factors such as species, DBH, tree size, condition, and structural defects, although the importance of these factors varies across species, storms, and locations. Kabir et al. (2018) also showed that routine characteristics such as species and DBH can contain useful predictive information. [1], [2], [15]

Theme 2

Expanding on visible structural defect parameters, research on visible structural defects show a less consistent picture. Although they are widely used as a factor in tree-risk assessments, studies disagree on how reliably distinct defects are associated with storm failure as some deterioration-related defects demonstrate useful relationships and other visually concerning structural forms perform less consistently. For instance, the 2020 study by Koeser et al. compared pre-storm assessed trees with post-storm results, and found that dead branches, deep planting, severe stem-girdling roots, and previous wounding were significantly associated with higher failure likelihood, while Nelson et al. (2022) state that weak branch unions were considerably less problematic, and some defects such as decay and dead wood corresponded positively with increased failure. Later in 2023 Koeser et al. reported that 14 trees had branch failure after hurricane Ian but none of these failures occurred at the defect that had originally been identified as the primary condition of concern. Therefore, the defect information usefulness is not homogeneous as its value differs with the type of defect and context. [4], [5], [7]

Nevertheless, Koeser et al. (2017) study adds that more detailed assessment Level 2 and Level 3 changed estimation of likelihood-of-failures ratings in comparison with Level 1 estimation ratings, implying that more information changes decisions made on tree-risk assessments. This makes the current research question about whether adding visible structural-defect information improves pre-storm prioritization worth studying. Although previous works have shown some association between defects and failure, and some have not, this still does not make the question about how much the full set of detailed defect observations improves prioritization beyond routine inventory data. [3]

Theme 3

The same group of studies has tested whether pre-storm visual assessments can actually identify trees that later fail during severe weather. Previously, Koeser et al. (2017) examined how assessment detail affected professional likelihood-of-failure ratings but did not validate those ratings against a later storm outcome. Later, Koeser et al. (2020) wanted to know whether arborists' pre-storm likelihood-of-failure ratings corresponded with actual later storm impact. Revisiting 2,069 trees on five properties after Tropical Storm Matthew, they recorded if each remained intact or experienced partial or whole-tree failure and compared these outcomes with the earlier risk assessments. The key results were that 94.1% of "Imminent" (failure expected very soon, even without severe weather), 38.8% of "Probable" (failure likely under normal weather conditions), 15.3% of "Possible" (failure could occur, especially under severe weather), and 0% of "Improbable" (failure unlikely during the specified assessment period) predicted tree sets were damaged. Additionally, they also conducted logistic-regression analysis, which identified parameters that were significantly associated with tree failure: species, DBH, and defects. [3], [4]

Another one of the studies that recorded both pre-storm and post-storm data is Nelson et al. (2022). Approximately 2,394 Tampa trees had information recorded before Hurricane Irma and were classified into ratings: Low, Moderate, High, and Extremely high. According to the results the professional inspection had some predictive signal as 28% and 28% of trees rated Extremely High were branch failure and whole-tree failure respectively. Yet, although High and Moderate trees had different results in branch failure percentage, their whole-tree failure were identical, which suggests that difference between those two categories did not translate into a difference in observed whole-tree failure. [7]

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Lastly, a 2026 study by McBride et al., the closest study about the effectiveness of limited visual assessment in urban tree storm failures prediction, similarly conducted pre- and post-storm assessments after the derecho wind disaster. During the pre-storm assessment only 38 out of 2,253 trees received Level 2 inspection based on Level 1 limited visual assessments. Strikingly, 169 (8%) of the trees were damaged, 67 (3%) underwent complete tree failure, and among 38 selected trees for Level 2 only 10 (26%) were damaged and 4 (11%) completely failed. In other words, approximately 94% of all subsequently damaged trees were outside that small elevated-risk tree set. Regarding the Level 2 inspected trees, among the 38 Level 2-assessed trees, 4 ultimately experienced whole-tree failure, and all 4 of those trees had previously been rated low-risk. [6]

Theme 4

The studies collectively demonstrate that storm-related tree failure remains difficult to predict consistently. Assessments contain useful signals, but their ability to identify which individual trees will subsequently fail varies substantially and can be poor even with strong overall classification performance. For example, McBride et al., identified a subgroup with 26.3% damage rate, but this subgroup contained only around 5.9% of all subsequently damaged trees; Nelson et al. determined identical 7% whole-tree failure rates among moderate- and high-risk trees, whereas Koeser et al. found much stronger separation across likelihood categories. [4], [6], [7]

Nevertheless, among the studies reviewed, the extent to which adding visible structural-defect information to routine inventory data improves urban-tree prioritization for the same trees and under the same inspection-capacity has not been clearly quantified.

Methodology

This study takes a post-positivist quantitative approach, using real-world numerical data to examine the relationship between tree characteristics and recorded storm damage while recognizing that the datasets may not capture every physical tree failure. A quantitative methodology will be used to compare the prioritization of routine inventory information with the same information plus visible structural-defect data. Two models A and B are to be compared based on what proportion of all later damaged trees were captured within the highest risk 5%, 10%, and 20% of ranked trees. This thresholds are used to represent a situation where a city can prioritize only a limited share of trees before a storm due to a low inspection capacity.

The study's target population is NYC-managed urban trees that had an eligible Level 2 assessment before Tropical Storm Isaias. After the final eligibility checks and data cleaning, the V4 analytical cohort contained 162,470 trees, including 1,965 linked verified storm failures.

Inclusion criteria: 1) identifiable tree; 2) completed Level 2 inspection; 3) inspection completed before August 4, 2020; 4) latest eligible inspection within 24 months; 5) linked risk assessment information; 6) trees still present when Isaias occurred.

Exclusions: 1) trees confirmed by Work Orders as removed before August 4, 2020; 2) ambigious Isaias outcomes; 3) records lacking reliable tree linkage; 4) records contaminated by storm-family predictor leakage.

These criteria make the positive class more specific and prevent ambiguous storm-related cases from being incorrectly treated as negative cases.

The study uses five NYC Parks datasets, where each one contributes different type of information: 1) Service Requests (Isaias-related damage), 2) Inspections (eligible pre-storm Level 2 assessments), 3) Risk Assessments (visible structural-defect information), 4) Tree Points (tree identity/GlobalID and species), 5) Work Orders (trees removed before storm). These data are linked into one tree with both pre- and post-storm information. Preliminary testing showed that linking Service Requests through only their own ID connected about 45% of strong storm requests but allowing own or parent ID increased linkage to about 93%. Linked inspections are then connected to individual tree IDs and their corresponding Risk Assessments. [10]-[14]

The outcome is coded as y=1 when tree is linked to a Hazard Service Request and indicates strong failure category (Tree Down, Tree Split, and other) during the primary Isaias period (August 4 00:00 to August 7 00:00, 2020); request has verified operational resolution; it is linked to an identifiable tree. y=0 indicates no recorded and verified storm induced failure report and no

ambiguous storm-related damage record during the exclusion period. Uncertain damage such as Hanging Limb, Tree Leaning are excluded to reduce outcome misclassification. [12]

Component	Model A: Routine inventory	Model B: Routine + defects
Tree population	Same eligible trees	Same eligible trees
Routine variables	Species/genus, DBH, condition, structure, inspection age, reactive/non-reactive status, geographic adjustment if used	Same variables as Model A
Structural-defect variables	Not included	Dead wood, decay, cracks, weak branch unions, root problems, poor tree architecture, cankers, defect locations, etc.
Outcome	Recorded verified Isaias storm damage	Same
Algorithm	Logistic regression	Logistic regression
Validation	Same held-out/spatial validation groups	Same held-out/spatial validation groups
Purpose	Baseline prioritization using routine information	Test whether adding visible defect information improves prioritization

Logistic regression is used for both models as the outcome is binary and the aim is to compare information sets. Trees are split for training and held-out validation groups as models should be tested on trees they did not learn from. Same validation trees are going to be used for both models. Because nearby trees may share similar storm exposure spatial grouped validation is preferred. Each model provides every validation tree a predicted risk probability. Top 5%, 10%, and 20% will be checked to represent different inspection capacities and the main metric will be Recall@10%, a proportion of all storm-damaged trees captured inside the top 10%. Model B Recall@10% and Model A Recall@10% will be compared and the difference will show added prioritization value of visible structural-defect information.

Limitations

There are several limitations to this study. Administrative records may miss unreported failures. Level 2 assessed trees may not represent all NYC trees. Only one city and one storm are analyzed in this study, which limits its results from being generalized. Inspections might be up to 24 months old. And there might still be some linkage uncertainty remaining.

Expected Outcomes

- if Model B captures more later-damaged trees at the same 5/10/20% capacity: defect information adds operational value;
- if improvement is small or absent: collecting detailed defect information may add limited prioritization benefit beyond routine inventory data;

- either result is useful because it quantifies the value of collecting detailed defect information rather than assuming it is beneficial.

Timeline

- Week 1: Finalize the analytical cohort, complete data linkage, remove trees that were already removed before Isaias, and confirm the final outcome definitions.
- Week 2: Prepare the variables for Model A and Model B, clean the datasets, and fit both logistic-regression models.
- Week 3: Validate both models on held-out trees, rank trees by predicted risk, and compare Recall@5%, Recall@10%, and Recall@20%.
- Week 4: Conduct sensitivity analyses, interpret the difference between Models A and B, discuss limitations, and finalize the research report.

Conclusion

Overall, this study will test whether adding visible structural-defect information to routine inventory data meaningfully improves pre-storm tree prioritization under limited inspection capacity. By comparing two models on the same trees and evaluating how many later-damaged trees are captured within fixed inspection thresholds, the study aims to quantify the practical value of collecting more detailed defect information. The findings may help cities decide whether additional tree-risk assessments provide enough benefit to justify the extra time and resources required.

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